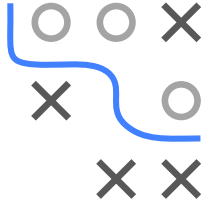
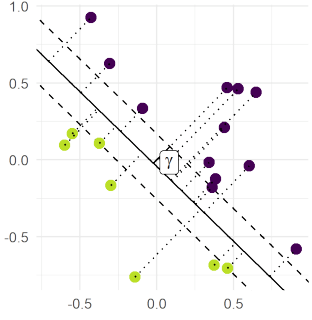
Supervised Learning



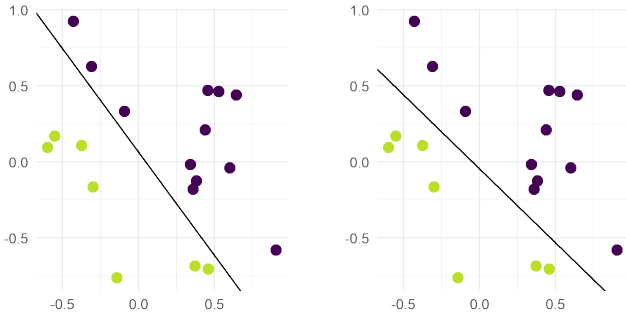
Linear Support Vector Machines
Linear Hard Margin SVM

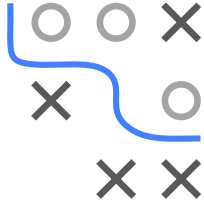


Learning goals

- Know that the hard-margin SVM maximizes the margin between data points and hyperplane
- Know that this is a quadratic program
- Know that support vectors are the data points closest to the separating hyperplane

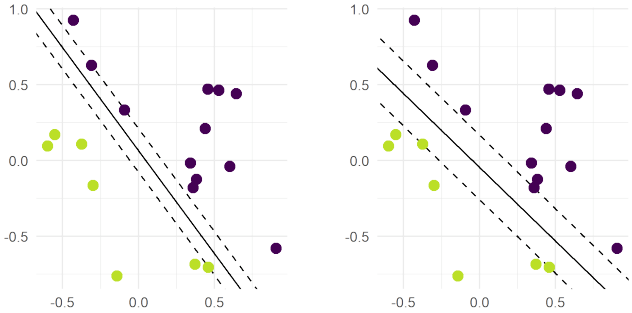
LINEAR CLASSIFIERS I

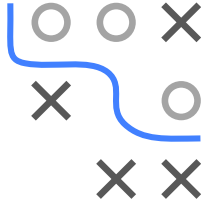




- We want study how to build a binary, linear classifier from solid geometrical principles.
- Which of these two classifiers is “better”?

LINEAR CLASSIFIERS II

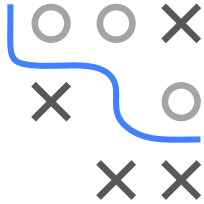




- We want study how to build a binary, linear classifier from solid geometrical principles.
- Which of these two classifiers is “better”?

→ The decision boundary on the right has a larger **safety margin.**

SUPPORT VECTOR MACHINES: GEOMETRY I



For labeled data $\mathcal{D} = \left(\left(\boldsymbol{x}^{(1)}, y^{(1)} \right), \ldots, \left(\boldsymbol{x}^{(n)}, y^{(n)} \right) \right)$, with $y^{(i)} \in \{-1, +1\}$:

- Assume linear separation by $f(\boldsymbol{x}) = \boldsymbol{\theta}^\top \boldsymbol{x} + \theta_0$, such that all $+$-observations are in the positive halfspace

$$\{\boldsymbol{x} \in \mathcal{X} : f(\boldsymbol{x}) > 0\}$$

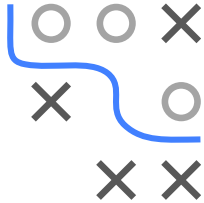
  and all $-$-observations are in the negative halfspace

$$\{\boldsymbol{x} \in \mathcal{X} : f(\boldsymbol{x}) < 0\}.$$

- For a linear separating hyperplane, we have

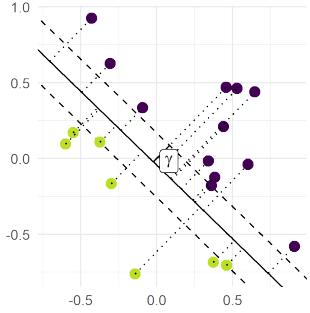
$$y^{(i)} \underbrace{\left(\boldsymbol{\theta}^\top \boldsymbol{x}^{(i)} + \theta_0 \right)}_{= f\left(\boldsymbol{x}^{(i)}\right)} > 0 \quad \forall i \in \{1, 2, ..., n\}.$$

SUPPORT VECTOR MACHINES: GEOMETRY II

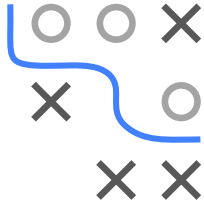


- $$d\left(f, \boldsymbol{x}^{(i)}\right) = \frac{y^{(i)} f\left(\boldsymbol{x}^{(i)}\right)}{\|\boldsymbol{\theta}\|} = y^{(i)} \frac{\boldsymbol{\theta}^T \boldsymbol{x}^{(i)} + \theta_0}{\|\boldsymbol{\theta}\|}$$

  computes the (signed) distance to the separating hyperplane $f(\boldsymbol{x}) = 0$, positive for correct classifications, negative for incorrect.
- This expression becomes negative for misclassified points.



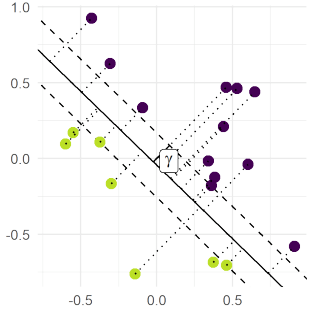
SUPPORT VECTOR MACHINES: GEOMETRY III



- The distance of f to the whole dataset $\mathcal{D}$ is the smallest distance

$$\gamma = \min_i \left\{ d\left(f, \boldsymbol{x}^{(i)}\right) \right\}.$$

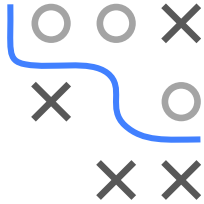
- This represents the “safety margin”, it is positive if f separates and we want to maximize it.



MAXIMUM MARGIN SEPARATION I

We formulate the desired property of a large “safety margin” as an optimization problem:

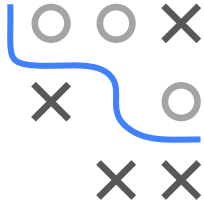
$$\begin{aligned} \max_{\boldsymbol{\theta},\theta_0} \quad & \gamma \\ \text{s.t.} \quad & d\left(f, \boldsymbol{x}^{(i)}\right) \geq \gamma \quad \forall\, i \in \{1, \ldots, n\}. \end{aligned}$$



- The constraints mean: We require that any instance i should have a “safety” distance of at least γ from the decision boundary defined by $f(= \boldsymbol{\theta}^T \boldsymbol{x} + \theta_0) = 0$.
- Our objective is to maximize the “safety” distance.

MAXIMUM MARGIN SEPARATION I

We reformulate the problem:

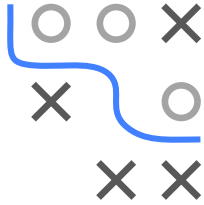


$$\begin{aligned} \max_{\boldsymbol{\theta},\theta_0} \quad & \gamma \\ \text{s.t.} \quad & \frac{y^{(i)}\left(\left\langle \boldsymbol{\theta}, \boldsymbol{x}^{(i)} \right\rangle + \theta_0\right)}{\|\boldsymbol{\theta}\|} \geq \gamma \quad \forall\, i \in \{1, \ldots, n\}. \end{aligned}$$

- The inequality is rearranged by multiplying both sides with $\|\boldsymbol{\theta}\|$:

$$\begin{aligned} \max_{\boldsymbol{\theta},\theta_0} \quad & \gamma \\ \text{s.t.} \quad & y^{(i)}\left(\left\langle \boldsymbol{\theta}, \boldsymbol{x}^{(i)} \right\rangle + \theta_0\right) \geq \|\boldsymbol{\theta}\|\gamma \quad \forall\, i \in \{1, \ldots, n\}. \end{aligned}$$

MAXIMUM MARGIN SEPARATION II



- Note that the same hyperplane does not have a unique representation:

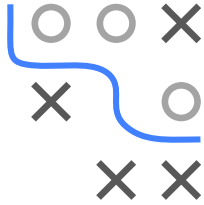
$$\{\boldsymbol{x} \in \mathcal{X} \mid \boldsymbol{\theta}^\top \boldsymbol{x} = 0\} = \{\boldsymbol{x} \in \mathcal{X} \mid c \cdot \boldsymbol{\theta}^\top \boldsymbol{x} = 0\}$$

  for arbitrary $c \neq 0$.

- To ensure uniqueness of the solution, we make a reference choice – we only consider hyperplanes with $\|\boldsymbol{\theta}\| = 1/\gamma$:

$$\begin{aligned} \max_{\boldsymbol{\theta}, \theta_0} \quad & \gamma \\ \text{s.t.} \quad & y^{(i)} \left(\left\langle \boldsymbol{\theta}, \boldsymbol{x}^{(i)} \right\rangle + \theta_0 \right) \geq 1 \quad \forall\, i \in \{1, \ldots, n\}. \end{aligned}$$

MAXIMUM MARGIN SEPARATION III



- Substituting $\gamma = 1/\|\boldsymbol{\theta}\|$ in the objective yields:

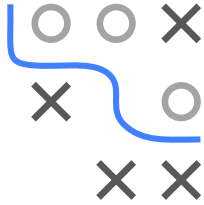
$$\begin{aligned} \max_{\boldsymbol{\theta},\theta_0} \quad & \frac{1}{\|\boldsymbol{\theta}\|} \\ \text{s.t.} \quad & y^{(i)}\left(\left\langle \boldsymbol{\theta}, \boldsymbol{x}^{(i)} \right\rangle + \theta_0\right) \geq 1 \quad \forall\, i \in \{1, \ldots, n\}. \end{aligned}$$

- Maximizing $1/\|\boldsymbol{\theta}\|$ is the same as minimizing $\|\boldsymbol{\theta}\|$, which is the same as minimizing $\frac{1}{2}\|\boldsymbol{\theta}\|^2$:

$$\begin{aligned} \min_{\boldsymbol{\theta},\theta_0} \quad & \frac{1}{2}\|\boldsymbol{\theta}\|^2 \\ \text{s.t.} \quad & y^{(i)}\left(\left\langle \boldsymbol{\theta}, \boldsymbol{x}^{(i)} \right\rangle + \theta_0\right) \geq 1 \quad \forall\, i \in \{1, \ldots, n\}. \end{aligned}$$

QUADRATIC PROGRAM I

We derived the following optimization problem:



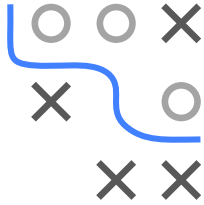
$$\begin{aligned} \min_{\boldsymbol{\theta},\theta_0} \quad & \frac{1}{2}\|\boldsymbol{\theta}\|^2 \\ \text{s.t.} \quad & y^{(i)}\left(\left\langle \boldsymbol{\theta}, \boldsymbol{x}^{(i)} \right\rangle + \theta_0\right) \geq 1 \quad \forall\, i \in \{1, \ldots, n\}. \end{aligned}$$

This turns out to be a **convex optimization problem** – particularly, a **quadratic program**: The objective function is quadratic, and the constraints are linear inequalities.

This is called the **primal** problem. We will later show that we can also derive a dual problem from it.

We will call this the **linear hard-margin SVM**.

SUPPORT VECTORS I



- Some $(\boldsymbol{x}^{(i)}, y^{(i)})$ will have minimal margin, $y^{(i)} f\left(\boldsymbol{x}^{(i)}\right) = 1$, fulfilling the inequality constraints with equality.
- Implies a distance of $\gamma = 1/\|\boldsymbol{\theta}\|$ from separating hyperplane.
- Geometrically obvious that optimal hyperplane doesn't depend on observations with larger distance.
- Hence, we call some of these minimal margin vectors (but not necessarily all) support vectors.
- More formal definition: in upcoming section on duality.

